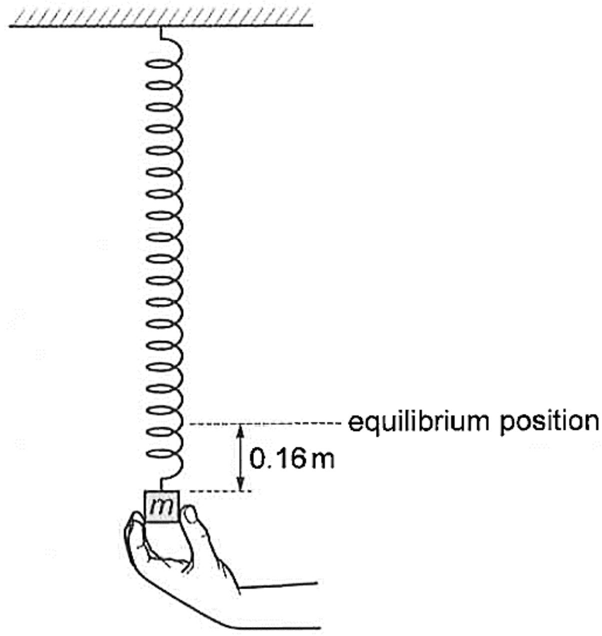
4	A mass m is suspended from a vertical spring of spring constant k attached to a fixed support. The mass is pulled down and held at a vertical displacement of 0.16 m from its equilibrium position, as shown in Fig. 4.1.  **Fig. 4.1** The mass is released.		
	(a)	Show that the mass's acceleration a is related to its displacement x from the equilibrium position by the equation: $$a = -\frac{k}{m}x.$$ Explain your working.	
			[3]
	L3	**Presentation 1:** **At Equilibrium position, no net force.** **upward spring force = downward weight** **At displacement x (downwards) below the Equilibrium position,** **increase in spring extension by x increases the upward spring force by kx,** **while the downward weight is unchanged.** **Thus,** **Net force is kx upwards.** **Take direction downwards as positive. Then displacement (x) is positive while net force (kx) is negative.** **By Newton's 2nd law of motion,**	

		Net force = ma **-kx = ma** $a = -\frac{k}{m}x$ **[Shown]** **<u>Presentation 2:</u>** **At Equilibrium position, no net force.** **Spring force = Weight** **$kx_o = mg$ where x_o: spring extension at Equilibrium position —(1)** **Take direction downwards as positive.** **At displacement x (downwards) below the Equilibrium position, spring extension is now (x_o + x), and net force is non-zero.** **Net force = - (New Spring force) + Weight** **= - k(x_o + x) + mg —(2)** **Sub (1) into (2): Net force = - k(x_o + x) + kx_o = -kx** **By Newton's 2nd law of motion,** **Net force = ma** **-kx = ma** $a = -\frac{k}{m}x$ **[Shown]** **Marks scheme:** **1 mark:** use of Hooke's Law, F = kx **1 mark:** how Negative sign arise is clear from sign convention **1 mark:** use of Newton's 2nd law	 **M1** **M1** **M1** **A0**
	(b)	The mass undergoes simple harmonic oscillations described by the equation in **(a)**. Show that the period T of the oscillations of the mass is given by: $$T = 2\pi\sqrt{\frac{m}{k}}$$	
			[2]
	L2	**Compare with the SHM equation: a = -ω^2x** $\omega^2 = \frac{k}{m}$ $\left(\frac{2\pi}{T}\right)^2 = \frac{k}{m}$ $T = 2\pi\sqrt{\frac{m}{k}}$ **[Shown]**	**M1** **M1** **A0**
	(c)	Ten oscillations are timed using a stopwatch. The data for the mass and the time, together with their uncertainties, are shown in Table 4.1.	

Table 4.1

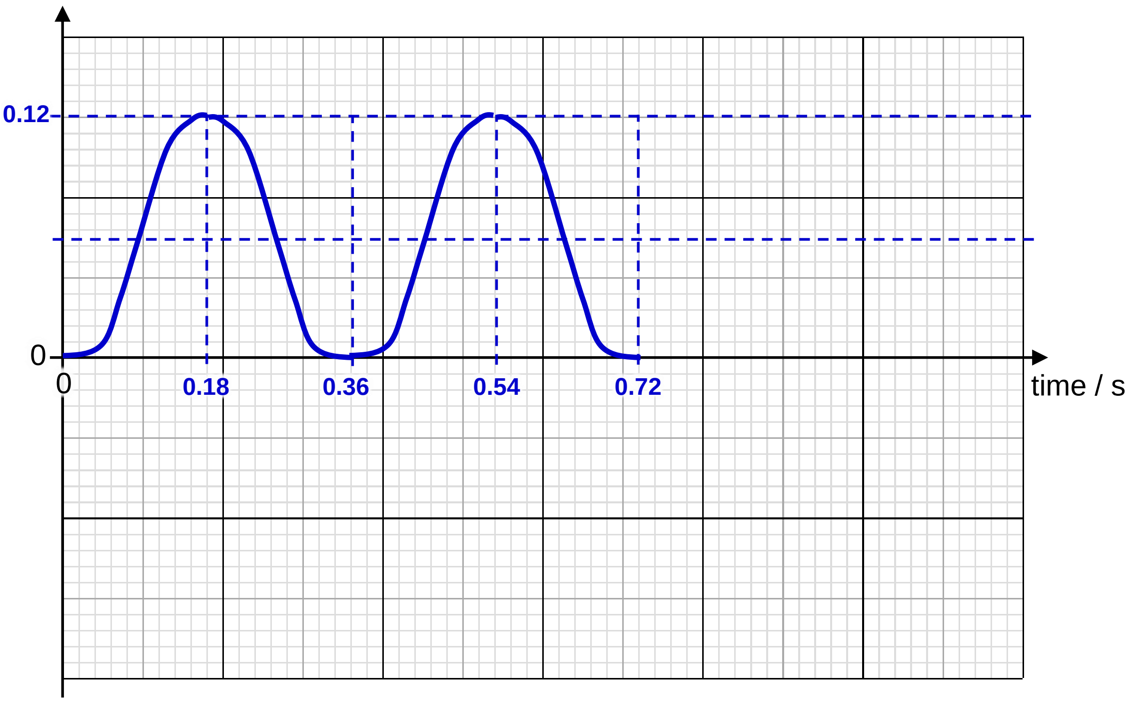
time for 10 oscillations / s	7.2 $\pm$ 0.2
m / g	120 $\pm$ 1%

		Determine the value of *k* together with its actual uncertainty. Give your answer to an appropriate number of significant figures.	
		k = ... ± .... N m^{-1}	[3]
	L2	$T = \frac{t}{10}$ T = $\frac{7.2\ s}{10}$ = 0.72 s $\frac{\Delta T}{T} = \frac{\Delta t}{t} = \frac{0.2}{7.2}$ Making k the subject in the equation from (b), $k = \frac{4\pi^2 m}{T^2} = \frac{4\pi^2(0.120\ kg)}{(0.72\ s)^2}$ = **9.1385 N m^{-1}** $\frac{\Delta k}{k} = \frac{\Delta m}{m} + 2\frac{\Delta T}{T} = \frac{1}{100} + 2\left(\frac{0.2}{7.2}\right)$ = 0.065556 Δk = 0.065556 k = 0.065556 (9.1385) = **0.6 N m^{-1}** **(1 sig. fig.)** **(k ± Δk) = (9.1 ± 0.6) N m^{-1}** **(Round off k to the same precision as Δk)**	 **M1** **M1** **A1**
	(d)	Calculate the total energy of oscillations of the spring-mass system.	
		total energy = ... J	[2]
	L2	**Total energy of oscillations = Maximum KE** = ½ m v_o^2 = ½ m $(\omega x_o)^2$ = ½ m$\omega^2 x_o^2$ = ½ (0.120 kg) $\left(\frac{2\pi}{0.72\ s}\right)^2$ (0.16 m)^2 = 0.11697 = **0.12 J**	 **C1** **A1**
	(e)	On Fig. 4.2, sketch a graph to show the variation with time of the kinetic energy of the mass for one complete oscillation, starting from the time of release. Label the axes with values obtained from **(c)** and **(d)**.	

Fig. 4.2

[2]

L2



1 mark for: • Initial conditon: start from (0, 0) since at time = 0, KE = 0. • Shape: sine-squared graph; symmetrical.	B1
1 mark for: • Axes labelled with values found earlier. • Drawn for one complete oscillation.	B1

[Total: 12]

5 Coherent light is incident normally on a double slit, as shown in Fig. 5.1.